

B.TECH. (CSE/CS/CE)
SEVENTH SEMESTER (DETAILED SYLLABUS)

Artificial Intelligence (BCS701)		
Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course, the student will be able to understand		
CO 1	Understand the fundamentals of Artificial Intelligence, intelligent agents, and various approaches to problem-solving in AI.	K2
CO 2	Apply uninformed and informed search strategies, heuristics, and optimization techniques to solve classical AI problems and games.	K3
CO 3	implement logical reasoning techniques using propositional and first-order logic, including inference strategies and knowledge representation methods.	K4
CO 4	Analyze uncertainty in knowledge representation using probabilistic reasoning, fuzzy logic, and basic neural network concepts.	K4
CO 5	Evaluate and demonstrate AI applications in areas such as natural language processing (NLP), robotics, multi-agent systems, and Explainable AI (XAI) through real-world problem examples.	K5
DETAILED SYLLABUS		3-0-0
Unit	Topic	Proposed Lecture
I	Introduction to Artificial Intelligence & Intelligent Agents: Definition and scope of AI, History and applications of AI, Characteristics of Intelligent Agents, Types of agents and environments, Agent architecture, Problem Solving Approach to Typical AI problems, Problem-solving agents. Example problems and approaches.	08
II	Problem Solving & Search Strategies: Uninformed Search Strategies: BFS, DFS, Iterative Deepening, Informed Search Strategies: Greedy Best-First Search, A* Search, Heuristics and Optimization, Hill Climbing, Simulated Annealing, Constraint Satisfaction Problems, Game Playing: Min-max, Alpha-Beta Pruning, Stochastic & Partially Observable Games.	08
III	Knowledge Representation & Reasoning: Propositional and First Order Logic, Syntax, Semantics, and Inference, Knowledge-based agents: Wumpus world, Logic Programming using Prolog, Forward and Backward Chaining, Resolution, Ontological Engineering and Reasoning.	08
IV	Uncertainty & Learning Techniques: Introduction to uncertainty and probabilistic reasoning, Bayes' Rule, Bayesian Networks, Fuzzy logic and handling imprecision, Neural Networks (basics only): Perceptron, Backpropagation (intro level), Fundamentals of Machine Learning in AI context, Introduction to supervised and unsupervised learning.	08
V	Applications of AI & Multi-Agent Systems: Natural Language Processing, Machine Translation, Information Retrieval and Extraction, Robotics: Perception, Planning, and Motion, Speech Recognition, Software Agents: Architecture, Communication, Trust, Multi-agent Negotiation and Reputation. Explainable AI (XAI) – Importance of interpretability, techniques for explaining black-box models, trust in AI, case studies in NLP and vision.	08
Recommended Textbooks:		
1. Stuart Russell & Peter Norvig, <i>Artificial Intelligence: A Modern Approach</i>, 4th Edition, Pearson, 2022 2. Ivan Bratko, <i>Prolog: Programming for Artificial Intelligence</i>, 4th Edition, Addison-Wesley 3. Nils J. Nilsson, <i>The Quest for Artificial Intelligence</i>, Cambridge University Press 4. David Poole & Alan Mackworth, <i>Artificial Intelligence: Foundations of Computational Agents</i>, Cambridge Press		

Internet of Things (BCS070)		
Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course, the student will be able to understand		
CO 1	Describe the basic concepts, principles and challenges in IoT.	K2
CO 2	Illustrate functioning of hardware devices and sensors used for IoT.	K2
CO 3	Analyze network communication aspects and protocols used in IoT.	K4
CO 4	Apply IoT for developing real life applications using Arduinio programming.	K3
CP 5	To develop IoT infrastructure for popular applications	K2, K3
DETAILED SYLLABUS		3-1-0
Unit	Topic	Proposed Lecture
I	Internet of Things (IoT): Vision, Definition, Conceptual Framework, Architectural view, technology behind IoT, Sources of the IoT, M2M Communication, IoT Examples. Design Principles for Connected Devices: IoT/M2M systems layers and design standardization, communication technologies, data enrichment and consolidation, ease of designing and affordability	08
II	Hardware for IoT: Sensors, Digital sensors, actuators, radio frequency identification (RFID) technology, wireless sensor networks, participatory sensing technology. Embedded Platforms for IoT: Embedded computing basics, Overview of IOT supported Hardware platforms such as Arduino, NetArduino, Raspberry pi, Beagle Bone, Intel Galileo boards and ARM cortex.	08
III	Network & Communication aspects in IoT: Wireless Medium access issues, MAC protocol survey, Survey routing protocols, Sensor deployment & Node discovery, Data aggregation & dissemination	08
IV	Programming the Arduinio: Arduinio Platform Boards Anatomy, Arduinio IDE, coding, using emulator, using libraries, additions in arduinio, programming the arduinio for IoT.	08
V	Challenges in IoT Design challenges: Development Challenges, Security Challenges, Other challenges IoT Applications: Smart Metering, E-health, City Automation, Automotive Applications, home automation, smart cards, communicating data with H/W units, mobiles, tablets, Designing of smart street lights in smart city.	08
Text books: 1. Olivier Hersent, David Boswarthick, Omar Elloumi "The Internet of Things key applications and protocols", wiley 2. Jeeva Jose, Internet of Things, Khanna Publishing House 3. Michael Miller "The Internet of Things" by Pearson 4. Raj Kamal "INTERNET OF THINGS", McGraw-Hill, 1ST Edition, 2016 5. Arshdeep Bahga, Vijay Madisetti "Internet of Things (A hands on approach)" 1ST edition, VPI publications, 2014 6. Adrian McEwen, Hakin Cassimally "Designing the Internet of Things" Wiley India		

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BOE070	PROJECT MANAGEMENT	3L 0T 0P	Credits: 3
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Course Objective: This course aims to equip students with essential knowledge and skills in project management, covering the complete project lifecycle—from initiation and planning to execution, control, and closure. It emphasizes practical tools like PERT/CPM, quality management, resource planning, and project finance for successful project implementation across diverse domains.

After completion of the course student will be able to:

COs	CO Statement	BL
CO1	Discuss complete structure of project management and analyze the scope of project planning.	K2
CO2	Identify different project selection methods.	K2
CO3	Explain the importance of procurement and its techniques.	K2
CO4	Define the guidelines required for project control and its controlling techniques.	K2
CO5	Outline the basic idea of projects and its initial management.	K2

Unit	Content	Hours
I	Introduction: Characteristics of a project types of projects, Project Management Body of Knowledge (PMBOK), role of project manager and his qualities, project organization and benefits, idea generation, needs of society, import substitution, project lifecycle, project charter, project sponsor. Project Planning: Customer needs, stake holder concept, project scope, feasibility study and report, baseline plan, SWOT analysis, project organization structure and hierarchy, project teams, formation, attitude and aptitude.	8
II	Structure: Project selection methods, breakeven analysis, DCF methods, project implementation, estimation, cost, price, value, scheduling, bar charts, network diagrams, PERT and CPM, schedule crashing, simple introduction to risk management, probability in project management, decision trees.	8
III	Procurement: Vendor selection methods, JIT, supply chains, quality, quality circles, quality control and quality assurance, cause and effect analysis, ISO and concepts of total quality management and six sigma, resource planning and allocation, availability and constraints of resources, resource leveling and crashing.	8

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IV	Project Control: Project scope, project change request, and control of schedule, resources, cost and quality, project communications, channels, means, meetings, project reports, project audits Project evaluation, project close-out reports, guidelines, audit reports, maintenance and shutdown projects, plant turn- around and brief introduction to replacement analysis	8
V	Projects: Contour maps, sitemaps, plant layout, suitability of project site, preparation of site, selection and leasing of construction equipment special considerations in selection and location of projects, safety, health, human and environment al factors, project finance, international projects, joint ventures, collaborations, impact of culture, implementation, and handing over of projects.	8

Text Books:

1. Kamaraj Ramakrishna, "Essentials of Project Management", PHI Learning, New Delhi, 2010.
2. Prasanna Chandra, "Projects – Planning, analysis, selection, implementation and review", Tata McGraw-Hill, New Delhi, 2010.
3. Chitkara, "Construction Project Management", Tata McGraw- Hill, New Delhi.
4. Harold Kerzner, "Project Management", Wiley, New York.
5. Iqbal Hussein, Electric and Hybrid Vehicles: Design Fundamentals, CRC Press, 2003.
6. Mehrdad Ehsani, Yimi Gao, Sebastian E. Gay, Ali Emadi, Modern Electric, Hybrid Electric and Fuel Cell Vehicles: Fundamentals, Theory and Design, CRC Press, 2004.